Indrani Sen

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Experienced Software Engineer with 6+ years of enterprise experience and a **recently completed MS in Computer Science (Dec 2025)** from Georgia Tech. Expert in architecting secure, scalable distributed systems using **Java/Spring Boot** and migrating legacy monoliths to distributed scalable architecture. Proven track record of optimizing high-throughput data pipelines and combining rigorous backend engineering with modern **AI/ML infrastructure**

EDUCATION & CERTIFICATION

Georgia Institute of Technology | MS, Computer Science	**Dec 2025**
Kalinga Institute of Industrial Technology | MS, Computer Application	May 2014
AWS Certified Solutions Architect - Associate	Pursuing

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Tata Consultancy Services | Dec 2014 – Jun 2021

- **Engineered a high-throughput** ingestion pipeline for energy data, migrating from legacy line-by-line inserts to DB2 Bulk Loading, reducing database write latency by **40%** for millions of records and eliminating **$200k** annual licensing costs
- **Architected the distributed microservices** backend using Spring Boot, replacing legacy monoliths and enabling scalable modules for multi-state utility providers
- **Architected secure payment platform** with RESTful services and JWT authentication, increasing quarterly revenue collection by **12%**
- **Implemented end-to-end distributed tracing** and immutable audit logging for the ingestion layer, ensuring **100% data lineage** and processing integrity for critical infrastructure compliance
- **Developed multi-threaded file parsers** for device data processing pipelines, streamlining data flow from acquisition to database loading, improving resource utilization by **14%**
- **Managed cross-functional teams** of **12+** engineers across **$5M+** project budgets, implementing Agile/Scrum methodologies for multi-state utility organizations
- **Automated reporting workflows** eliminating **300** manual hours weekly through process optimization and analytics automation

TECHNICAL SKILLS

Programming: Java, Python, C++, JavaScript
Frameworks: Spring, Spring Boot, JWT, OAuth, Hibernate, Spring Security, Angular, REST API, Lombok, Log4j, FastAPI
AI & Data Engineering: GenAI, RAG, Unstructured Data Processing, Vector Search Algorithms, LLM Integration
Cloud/DevOps: AWS (Lambda, S3, API Gateway, CloudWatch), Docker, GitLab CI, SOC 2 Compliance Principles
Databases: PostgreSQL, MySQL, Oracle, Db2
Tools: Git, Eclipse, VS Code, Postman, Swagger, JIRA, Maven
Testing & Quality: Unit Testing, Integration Testing, Selenium
Methodologies: Agile Development, Test-Driven Development

KEY TECHNICAL PROJECTS

Secure Full-Stack Analytics Platform (Java/Spring Boot) | Georgia Institute of Technology | Spring 2022

- **Designed** a scalable microservices-based platform using **Java Spring Boot** and **Angular**, implementing RESTful APIs for real-time data analytics
- **Secured** the application using **Spring Security** and **JWT (JSON Web Tokens)**, enforcing stateless authentication
- **Optimized** PostgreSQL database queries and indexing strategies, improving API response latency by **40%** while maintaining **98%-unit test coverage** via JUnit

Enterprise Unstructured Data Compliance Pipeline (Python/RAG) **|** Personal Project |Dec 2025

- **Architected** an end-to-end ingestion engine for unstructured data (PDF/Docx), utilizing modular document processors to normalize messy inputs for downstream AI agents
- **Implemented** a "Security-First" middleware using Regex/NER patterns to redact PII (SSN, Email) prior to inference, ensuring compliance with **SOC 2** privacy standards
- **Engineered** a self-verification loop with structured JSON audit logging, creating an immutable trail of AI decisions to prevent hallucinations and ensure processing integrity

GenAI-Powered Healthcare Risk Platform - Georgia Tech | 2025

- **Architected a provider-facing dashboard** using FastAPI (backend) and React (frontend), processing FHIR-compliant JSON patient data to calculate clinical risk scores
- **Integrated Google Gemini LLM** to transform Framingham risk assessments into personalized, evidence-based patient education and lifestyle recommendations
- **Managed end-to-end delivery** including Docker containerization for consistent deployment and the generation of synthetic datasets to test high-risk patient scenario**s**

PUBLICATIONS

"Path to Personalization: A Systematic Review of GenAI in Engineering Education" (KDD 2024)
Co-authored systematic review of 162 GenAI studies (KDD 2024); led global team of 6 researchers to define taxonomy and crowdsource content analysis